

RiNNAL-POP: A Low-rank ALM for Polyhedral-SDP and Moment-SOS Relaxations of Polynomial Optimization

Di Hou

National University of Singapore

Joint work with Tianyun Tang (UChicago), Kim-Chuan Toh (NUS)

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Polynomial Optimization

Polynomial Optimization

$$\min_{x \in \mathbb{R}^n} \left\{ f_0(x) \mid f_i(x) = 0, i \in [m], x \in D \right\}, \quad (\text{POP})$$

- ▶ $f_i(x)$: multivariate polynomials; D : semialgebraic domain.
- ▶ Inequalities can be encoded in D or by (squared) slack variables. (Ding and Wright, 2025)

Moment-SOS Relaxation

(J. B. Lasserre, 2001)

- ▶ Lower bound from SDP hierarchy.
- ▶ **Asymptotically** exact as order r increases.
- ▶ Var&Cons size grows rapidly: $\Omega(n^{2r})$.

Conic Reformulation

(Burer, 2009; Kim, Kojima, and Toh, 2020)

- ▶ Exact at a **fixed-order** τ .
- ▶ Untractable cone $\text{co}(K)$.
- ▶ Lower bound by Poly-SDP relaxations.

RiNNAL-POP can solve both (Mom-SOS) and (Poly-SDP) relaxations.

Conic Reformulation

Convex Conic Reformulation (Kim, Kojima, and Toh, 2020)

$$\min_X \left\{ \langle Q^0, X \rangle \mid \langle Q^i, X \rangle = 0, i \in [m], X_{11} = 1, X \in \text{co}(K) \right\}. \quad (\text{COP})$$

- Monomial degree:

$$\tau \geq \max \{ \lceil \deg f_i / 2 \rceil \mid i = 0, \dots, m \}.$$

- Monomial vector:

$$u_\tau(x) := (x^\alpha)_{|\alpha| \leq \tau} \in \mathbb{R}^{n_\tau}, \quad n_\tau = \binom{n+\tau}{\tau}.$$

- Rank-one lift:

$$X := u_\tau(x) u_\tau(x)^\top \in \mathbb{S}^{n_\tau}.$$

- Lifted set:

$$K := \left\{ u_\tau(x) u_\tau(x)^\top : x \in D \right\}.$$

	1	x_1	x_2	...	x^β
1	1	x_1	x_2	...	x^β
x_1	x_1	x_1^2	$x_1 x_2$	...	$x_1 x^\beta$
x_2	x_2	$x_2 x_1$	x_2^2	...	$x_2 x^\beta$
...	...	...	...	...	...
x^α	x^α	$x^\alpha x_1$	$x^\alpha x_2$	...	$x^{\alpha+\beta}$

Tractable OA of $\text{co}(K)$

Example of $\text{co}(K)$

$$\begin{aligned}\tau = 1, D = \mathbb{R}^n &\implies \text{co}(K) = \mathbb{S}_+^{n+1}, \\ \tau = 1, D = \mathbb{R}_+^n &\implies \text{co}(K) = \text{CPP}^{n+1}.\end{aligned}$$

Outer Approximation

$$\text{co}(K) \subseteq \mathbb{S}_+^{n_\tau} \cap \mathcal{C} \cap \mathcal{P}$$

PSD cone ($\mathbb{S}_+^{n_\tau}$)

$$X \succeq 0.$$

Consistency cone ($\mathcal{C}$)

$$\begin{aligned}X_{\alpha,\beta} &= X_{\gamma,\delta} \\ \text{if } \alpha + \beta &= \gamma + \delta\end{aligned}$$

Polyhedral cone ($\mathcal{P}$)

Problem dependent,
E.g., $D = \mathbb{R}_+^n \Rightarrow X \geq 0.$

This gives a tractable **SDP relaxation** and hence a lower bound for POP.

Identify Hidden Facial Structure

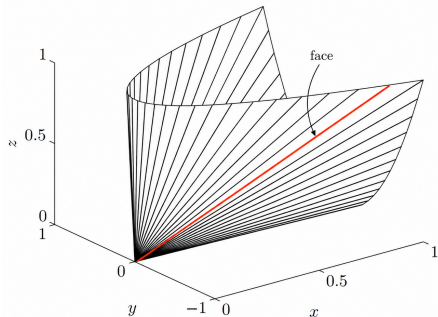
Key Insight. (COP) has many facial equalities.

Facial Constraints for $X \succeq 0$

$$AX = 0 \iff \langle A^\top A, X \rangle = 0$$

Sources.

- ▶ RLT (Sherali, 2007)
- ▶ Moment-SOS (J. B. Lasserre, 2001)



Example

$$a^\top x = b \implies \begin{bmatrix} -b & a^\top \end{bmatrix} \begin{bmatrix} 1 \\ x \end{bmatrix} \begin{bmatrix} 1 \\ x \end{bmatrix}^\top = 0, \implies \begin{bmatrix} -b & a^\top \end{bmatrix} X = 0.$$

Polyhedral-SDP Relaxation

Polyhedral-SDP Relaxation

$$\min_X \left\{ \langle Q^0, X \rangle \left| \begin{array}{l} \langle Q^i, X \rangle = 0, \forall i \in [\ell], \\ \textcolor{red}{AX} = 0, \quad X \in \mathbb{S}_+^{n_\tau}, \\ \textcolor{blue}{X}_{11} = 1, \quad X \in \mathcal{P} \cap \mathcal{C}, \end{array} \right. \right\}. \quad (\text{Poly-SDP})$$

- A unified framework that includes SDP/DNN/RLT/Mom-SOS relaxations.

Theorem 1 (Relaxation Tightness)

(Informal) Poly-SDP relaxation is as tight as moment-SOS relaxation.

RiNNAL-POP for solving (Poly-SDP)

Compact Reformulation

$$\begin{aligned} \min_{X,Y} \quad & \langle Q^0, X \rangle + \delta_{\mathcal{X}}(X) + \delta_{\mathcal{Y}}(Y) \\ \text{s.t.} \quad & X - Y = 0, \quad \mathcal{Q}(X) = 0. \end{aligned} \quad (\overline{\text{Poly-SDP}})$$

$$\mathcal{X} := \{X \in \mathbb{S}_+^{n^\tau} \mid AX = 0\}, \quad \mathcal{Y} := \{X \in \mathcal{P} \cap \mathcal{C} \mid X_{11} = 1\}, \quad \mathcal{Q}(X) := (\langle Q^i, X \rangle)_{i \in [\ell]}.$$

Bottleneck	Scale	RiNNAL-POP idea
Nonnegativity constraints	$\Omega(n^{2\tau})$	fast projection
Consistency constraints	$\Omega(n^{2\tau})$	fast projection
PSD variable	$\Omega(n^{2\tau})$	low-rank factorization $X = RR^\top$
Facial equalities	$\Omega(mn^\tau)$	$AX = 0 \iff AR = 0$
No Slater when $A \neq 0$	structural	work on the exposed face

Augmented Lagrangian Method

Augmented Lagrangian function.

$$\phi_k(X) = \langle Q^0, X \rangle + \frac{\sigma_k}{2} \|\mathcal{Q}(X) - \sigma_k^{-1} y^k\|^2 + \frac{\sigma_k}{2} \text{dist}^2(X - \sigma_k^{-1} W^k, \mathcal{Y}).$$

ALM Iteration

$$X^{k+1} \approx \arg \min_{X \in \mathcal{X}} \phi_k(X),$$

$$Y^{k+1} = \Pi_{\mathcal{Y}}(X^{k+1} - \sigma_k^{-1} W^k),$$

$$y^{k+1} = y^k - \sigma_k \mathcal{Q}(X^{k+1}),$$

$$W^{k+1} = W^k - \sigma_k (X^{k+1} - Y^{k+1}).$$

A Hybrid Method to Solve ALM Subproblem

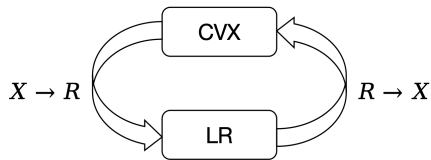


Figure: Two-phase transition

CVX Phase

Projected-gradient step on X

$$\begin{aligned} \min_X \quad & \phi(X) \\ \text{s.t.} \quad & AX = 0, X \in \mathbb{S}_+^{n_\tau}. \end{aligned}$$

- Primal feasibility enforcement.
- Global convergence (Lee et al., 2024).
- Automatic rank selection.

LR Phase

Projected-gradient step on R

$$\begin{aligned} \min_R \quad & \phi(RR^\top) \\ \text{s.t.} \quad & AR = 0. \end{aligned}$$

- Reduces variable dimension.
- Reduces the number of constraints.
- Reduces the objective value.

Automatic Rank Selection

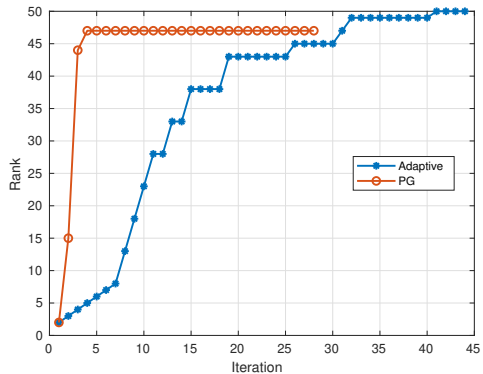


Figure: Initial rank: $r = 2$.

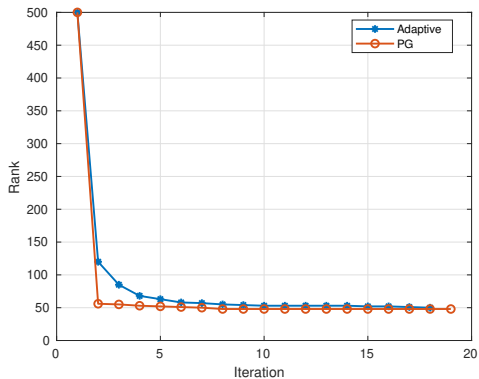


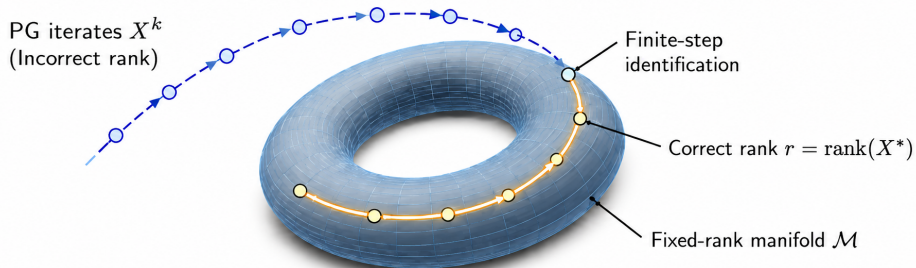
Figure: Initial rank: $r = n$.

Manifold Identification

Theorem 2 (Rank Identification)

After *finitely* many iterations,

$$\text{rank}(X^k) = \text{rank}(X^*) \quad \text{for all sufficiently large } k.$$

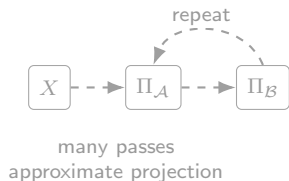


Exact Sequential Projection

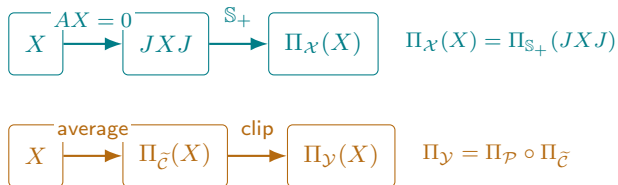
Projection Subproblems

$$\mathcal{X} = \{X \in \mathbb{S}_+^{n^\tau} \mid AX = 0\}, \quad \mathcal{Y} = \underbrace{\{X \in \mathbb{S}^{n^\tau} \mid X \geq 0\}}_{\mathcal{P}} \cap \underbrace{\{X \in \mathcal{C} \mid X_{11} = 1\}}_{\tilde{\mathcal{C}}}.$$

Generic alternating projection



Our structure: one pass is exact



- Exact one-pass projection.
- For $\Pi_{\mathcal{Y}}$: complexity reduced from $\mathcal{O}(n^{7\tau})$ to linear $\mathcal{O}(n^{2\tau})$.

Simple Verification of Global Optimality

Assume (Poly-SDP) admits an optimal solution satisfying the KKT conditions

$$\begin{aligned}AX &= 0, & \nabla\phi(X) - A^\top U - U^\top A &= S, \\ \langle X, S \rangle &= 0, & X \succeq 0, & \quad S \succeq 0.\end{aligned}$$

Theorem 3 (Recover dual variables)

A matrix $X \in \mathcal{X}$ is a minimizer of (Poly-SDP) if and only if

$$\begin{aligned}\widehat{S} &:= J(\nabla\phi(X))J \succeq 0 \\ X\widehat{S} &= 0\end{aligned}$$

Numerical Experiments

- ▶ Machine: MATLAB R2023b on an Intel Xeon E5-2680 v3 workstation with 96 GB RAM.
- ▶ Tolerance: relative KKT residue: $= 10^{-6}$; Maxtime: $= 3600s$.
- ▶ Baseline solvers:
 1. RiNNAL-POP (ours)
 2. SDPNAL+ (Yang, Sun, and Toh, 2015)
 3. GloptiPoly (Henrion, J.-B. Lasserre, and Löfberg, 2009)

Relaxation Comparison

Example: standard QP

$$\min \left\{ x^\top Q x : e^\top x = 1, x \in \mathbb{R}_+^n \right\}.$$

Order	1st	2nd
Poly-SDP-RLT	0	0
Mom-SOS	∞	0

Table: Relative gaps (%).

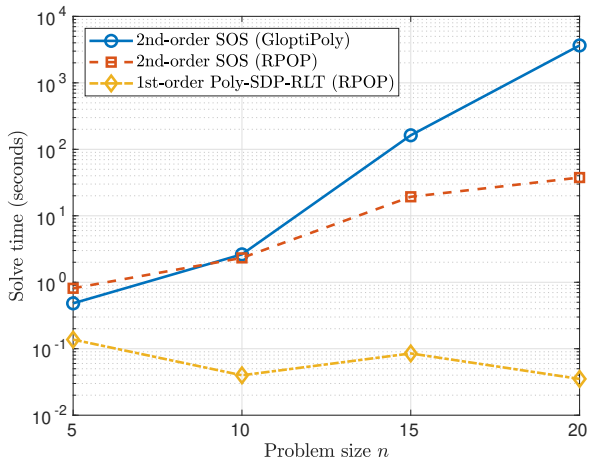


Figure: Computational time comparison.

Benchmark Problem Classes

Standard quadratic programming

$$\min \left\{ x^\top Q x : e^\top x = 1, x \in \mathbb{R}_+^n \right\}. \quad (\text{StQP})$$

Binary quadratic programs

$$\min \left\{ x^\top Q x + c^\top x \mid x \in \{0, 1\}^n \right\}. \quad (\text{BIQ})$$

Minimum bisection problem

$$\min \left\{ x^\top L x : e^\top x = \frac{n}{2}, x \in \{0, 1\}^n \right\}. \quad (\text{MBP})$$

Multiple quadratic knapsack problem

$$\max \left\{ x^\top Q x + c^\top x \mid A x \leq b, x \in \{0, 1\}^n \right\}. \quad (\text{MQKP})$$

Benchmark Problem Classes

Ball-constrained quartic minimization

$$\min \left\{ \langle c, [x]_4 \rangle \mid \|x\| \leq 1, x \in \mathbb{R}^{n-1} \right\}. \quad (\text{BQM})$$

Kurtosis-minimization portfolio problem

$$\min \left\{ \mathbb{E}_\xi \left[(x^\top (\xi - \mu))^4 \right] \mid \mu^\top x = \mu_0, x^\top \Sigma x = \sigma_0^2, e^\top x = 1, x \geq 0 \right\}. \quad (\text{KMP})$$

Copositivity of symmetric tensors

$$\min \left\{ \langle \mathcal{B}, \otimes^t x \rangle \mid e^\top x = 1, x \geq 0 \right\}. \quad (\text{CST})$$

Best nonnegative rank-one tensor approximation problem

$$\min \left\{ \|\mathcal{B} - \lambda x^{\otimes \alpha}\|^2 \mid \|x^{(i)}\| = 1, \lambda \geq 0, x^{(i)} \geq 0, \forall i \in [p] \right\}. \quad (\text{NTF})$$

Result Summary

Problem	n	Deg	Relax	Eq	Ineq	≥ 0	Speedup
(StQP)	1000 – 6000	2	1	Simplex	-	✓	34 – 36
(StQP)	20 – 80	2	2	Simplex	-	✓	3 – 6
(BIQ)	20 – 60	2	2	Binary	-	✓	10 – 25
(MBP)	30 – 60	2	2	Equipartition	-	✓	21 – 30
(MQKP)	20 – 40	2	2	Binary	Knapsack	✓	10 – 30
(BQM)	20 – 80	4	2	-	Ball	-	6 – 16
(KMP)	20 – 60	4	2	Mean–variance	-	✓	200 – 278
(CST)	1000 – 6000	2	1	Simplex	-	✓	18 – 500
(CST)	20 – 60	4	2	Simplex	-	✓	13 – 173
(CST)	5, 10	8	4	Simplex	-	✓	13 – 87
(NSTF)	20 – 80	3, 4	2	Spherical	-	✓	2 – 7
(NTF)	10 – 25	3, 4	2	Block spherical	-	✓	2 – 10

Conclusion

Problem	Relaxation	Solver
MBQP	SDP/DNN	RiNNAL
MBQP	SDP-RLT	RiNNAL+
POP	Poly-SDP	RiNNAL-POP

References:

Di Hou, Tianyun Tang, and Kim-Chuan Toh (2025a). “A low-rank augmented Lagrangian method for doubly nonnegative relaxations of mixed-binary quadratic programs”. In: *Operations Research* 74.3, pp. 1626–1647

Code: <https://github.com/HouDiOpt/RiNNAL>

Di Hou, Tianyun Tang, and Kim-Chuan Toh (2026). “RiNNAL+: a Riemannian ALM Solver for SDP-RLT Relaxations of Mixed-Binary Quadratic Programs”. In: *Mathematical Programming Computation*, pp. 1–57

Code: <https://github.com/HouDiOpt/RiNNALplus>

Di Hou, Tianyun Tang, and Kim-Chuan Toh (2025b). “A Low-rank Augmented Lagrangian Method for Polyhedral-SDP and Moment-SOS Relaxations of Polynomial Optimization”. In: *arXiv preprint arXiv:2512.06359*

Di Hou, Tianyun Tang, Kim-Chuan Toh, and Shiwei Wang (2026). “On the efficient computation of proximal operators of affine-constrained nonconvex functions”. In: *arXiv preprint arXiv:2602.23626*